

Problem-Solution Fit

Date	15 March 2026
Team ID	VBCUA-01
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure the proposed VBCUA solution accurately addresses the root causes, behaviors, and limitations of the target audience.

1. CUSTOMER SEGMENT(S) [CS] - Students giving oral concept explanations - Teachers evaluating communication and understanding - Academic project evaluators	6. CUSTOMER LIMITATIONS [CL] - Limited time for manual feedback - May have basic laptops/devices - No database or login required for current version	5. AVAILABLE SOLUTIONS [AS] - Manual viva: detailed but slow - Speech apps: fluency only - Quiz tools: concept check but not spoken explanation
2. PROBLEMS / PAINS [PR] - Hard to evaluate spoken answers consistently - Students do not know where they lose marks - Frequent issue during presentations and vivas	9. PROBLEM ROOT / CAUSE [RC] - Manual assessment lacks instant analytics - Concept accuracy and speech quality are checked separately - No simple report for improvement	7. BEHAVIOR [BE] - Students repeat practice without structured feedback - Teachers listen and judge manually - High intensity before exams, demos, and presentations
3. TRIGGERS TO ACT [TR] - Need to prepare for viva or presentation - Teacher asks for speaking improvement - Student wants concept-confidence check	10. YOUR SOLUTION [SL] - VBCUA records or uploads audio, transcribes it using Whisper, compares meaning using Sentence-BERT, analyses speech quality using Librosa, and generates a final score with PDF feedback.	8. CHANNELS OF BEHAVIOR [CH] ONLINE - Streamlit web app - GitHub project link - Render/VPS deployment link
4. EMOTIONS BEFORE / AFTER [EM] - Before: confused, nervous, unsure - After: informed, motivated, confident - Improves through clear suggestions		OFFLINE - Local laptop demo - Classroom evaluation - PDF report sharing